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Clostridioides difficile (including epidemiology)

Current practices to reduce the risk of transmission of *Clostridioides difficile* in France: results of a national survey

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ABSTRACT

Introduction: *Clostridioides difficile* is a major cause of healthcare-associated infections. In healthcare settings, interventions like contact precautions (CP), hand hygiene with soap and water, sporicidal environmental cleaning, and antimicrobial stewardship, are essential to reducing the risk *C. difficile* transmission. This study aimed to evaluate infection prevention and control (IPC) measures targeting *C. difficile* infection (CDI) in French hospitals.

Materials and methods: An anonymous online survey was conducted from August to December 2022. The assessed CDI-related IPC practices. Descriptive analysis was performed using IBM SPSS.

Results: A total of 350 professionals from all French regions completed the survey; 70.3 % were members of an infection control team members. CP for symptomatic patients were implemented by 70 %, and continued ≥48 h after diarrhea resolution of by 49 %; 4.9 % continued CP until discharge. Gloves, gowns, and aprons were worn before room entry by 62 %, 55 % and 21 % respectively. Hand hygiene with soap and water followed by alcohol-based hand rubs was reported by 85 %. Daily surface cleaning of rooms, toilets and floors was reported by 98 %, 96 %, and 87 % respectively. Cleaning was performed using sporicidal products (71 %), chlorine-containing agents (23 %), or non-sporicidal agents (5 %).

Conclusion: While key IPC measures are widely acknowledged, their implementation varies. Broader studies are needed to evaluate their effectiveness and address inconsistencies.

1. Introduction

By the mid-2000s, the frequency and severity of *Clostridioides difficile* (formerly known as *Clostridium difficile*) infection (CDI) has increased significantly worldwide, establishing it as a major cause of healthcare-associated infections [1]. The global burden of CDI continued to rise, particularly in countries with high to middle-income indices [2]. Based on its impact on healthcare systems, *C. difficile* has been classified as a pathogen of the highest priority for surveillance and epidemiological research [3,4].

C. difficile spores can persist on surfaces for prolonged periods, underscoring the critical role of the hospital environment in its the transmission [5]. Both asymptomatic and symptomatic patients serve as potential reservoir of *C. difficile* [6]. Furthermore, the hands of healthcare workers (HCW) are recognized as a primary vector for the transmission of *C. difficile* spore in clinical settings [7].

Infection prevention and control (IPC) measures, including contact precautions, hand hygiene using soap and water, environmental cleaning with sporicidal agents, and antimicrobial stewardship, are recommended by the European Society of Clinical Microbiology and Infectious Diseases (ESCMID) and by the Infectious Diseases Society of America and the Society for Healthcare Epidemiology of America (IDSA/SHEA)

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[8,9]. Nevertheless, the extent and consistency of IPC implementation can vary considerably between hospitals.

In this context, we aimed to assess the current landscape of CDI IPC practices across French hospitals.

2. Methods

As part of a European investigation conducted with the ESCMID Study Group for *C. difficile* (ESGCD) [10], in collaboration with the French prevention network for the prevention of healthcare associated infections (RéPIAS, renamed RéPIA in 2023 to include antimicrobial resistance), an anonymous online survey was conducted in France between August and December 2022 by the RéPIAS/RéPIA website (https://www.preventioninfection.fr) and throughout the RéPIAS newsletter. The questionnaire was structured around the recommendations issued by the ESCMID and IDSA/SHEA [8,9]. Data on specific practices implemented by hospitals are summarized in Fig. 1.

To assess adherence to CDI prevention guidelines, two scores were calculated [10]:

(1) ***Contact precautions (CP) score pending CDI diagnosis***: this included the application of CP for symptomatic patients while awaiting pending laboratory results, the use of disposable or dedicated patient care equipment, and isolation in a single room
(2) ***Global compliance score***: this composite score encompassed several core IPC practices: (i) implementation of CP pending diagnosis as outlined above, (ii) appropriate hand hygiene, (iii) daily surface cleaning with a sporicidal product, and (iv) continuation of CP for a at least 48 h after diarrhea resolution or until discharge. Each correct response was scored "1" and the incorrect answers "0." The total score (ranging from 0 to 7) was categorized as follows:
   a. Good (6–7 points; 80–100 %)
   b. Moderate (4–5 points; 50–79 %)
   c. Poor (<4 points; <50 %)

All components had to be correctly implemented for a hospital to be considered globally compliant. Data were analyzed using IBM SPSS 21.

3. Results

A total of 350 healthcare professionals participated in the survey, including 50 physicians or hospital pharmacists, 34 staff nurses, 76 nurses, 18 nursing assistants, 34 other healthcare staff, and 138 respondents who did not specify their role. Participants represented all regions of France (Fig. 2). The majority of participants reported being part of an infection control team (n = 245, 70 %) and working in public hospitals (n = 286, 82.6 %).

Table 1 summarizes the key findings. The use of personal protective equipment (apron, gloves, and gowns) before room entry varied considerably: aprons were used by 21.1 % of respondents, gloves by 54.9 %, and gowns by 62.3 %. Overall, 191 respondents (54.6 %) reported wearing gloves and either an apron or gown prior to entering the patient's room.

Hand hygiene using soap and water, followed by an alcohol-based hand rub, was reported by 85 % of participants. Daily cleaning of patient rooms was common: 97.7 % of respondents reported cleaning all horizontal and frequently touched surfaces, 95.7 % cleaned toilets, and 87.1 % cleaned floors. Cleaning was performed using sporicidal products by 71.4 % of respondents, chlorine-based agents by 23.4 %, and non-sporicidal products by 5.1 %. Daily cleaning of all surfaces, toilets and



Fig. 2. Geographical distribution of the healthcare institutions of professionals that took part in the national survey, France 2022.

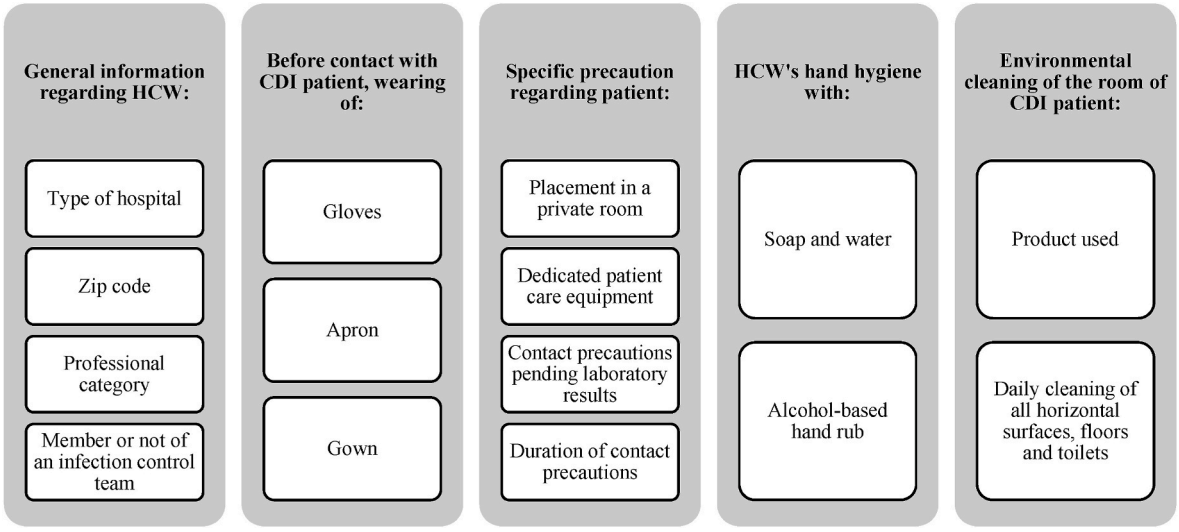


Fig. 1. Data collected regarding Infection prevention and control practices in French hospitals
CDI: *C. difficile* infection; HCW: health care worker.

Table 1
Description of practices related to CDI patients among participants of the national survey, France 2022.

	n (%)
Isolation precautions	
Contact Precautions if symptomatic pending laboratory results[a]	312 (89.1)
Use of disposable or dedicated patient care equipment[a]	59 (16.9)
Patient placement in a private room[a]	308 (88.0)
Precautions used before contact with CDI patient[b]	
Gloves before room entry[a]	218 (62.3)
Gloves only in case of an epidemic situation	2 (0.6)
Gloves only if direct personal care contact with the patient	140 (37.1)
Apron before room entry[a]	74 (21.1)
Apron only if direct personal care contact with the patient	118 (33.7)
Gown before room entry[a]	192 (54.9)
Gown only if direct personal care contact with the patient	105 (30)
Hand hygiene	
Soap and water	39 (11.1)
Soap and water followed by alcohol-based hand rub[a]	297 (84.9)
Alcohol-based hand rub	14 (4)
Environmental cleaning	
Daily cleaning of all horizontal and frequently touched surfaces, floors and toilets[a]	300 (85.7)
Daily cleaning of all horizontal and frequently touched surfaces in the room	342 (97.7)
Daily cleaning of room floor	305 (87.1)
Daily cleaning of toilet	335 (95.7)
Products used for environmental cleaning	
Chlorine-containing cleaning agents[a]	82 (23.4)
Other sporicidal agents[a]	250 (71.4)
Non-sporicidal agents	18 (5.1)

[a] Expected in accordance with current guidelines.
[b] Before contact with CDI patient, HCW wore gloves and either an apron or gown as recommended by guidelines.

floors with a sporicidal agent was reported by 291 participants (83.1 %).

CP for symptomatic patients were followed in 70 % (n = 246) of cases, and continued for at least 48 h after the resolution of diarrhea in 173 cases (49.0 %). However, 17 respondents (4.9 %) reported maintaining CP until hospital discharge (Fig. 3).

Regarding adherence to recommended IPC measures, 169 participants (48.3 %) demonstrated good compliance, 139 (39.7 %) moderate compliance, and 46 (12 %) poor compliance. Importantly, none respondent reported corrected implementation of all key components of the IPC guidelines. Among those with good adherence, participants were more likely to consistently perform adequate hand hygiene, follow the recommended duration of CP, and use sporicidal cleaning agents for environmental disinfection. However, the use of personal protective equipment before contact with CDI patients remained suboptimal.

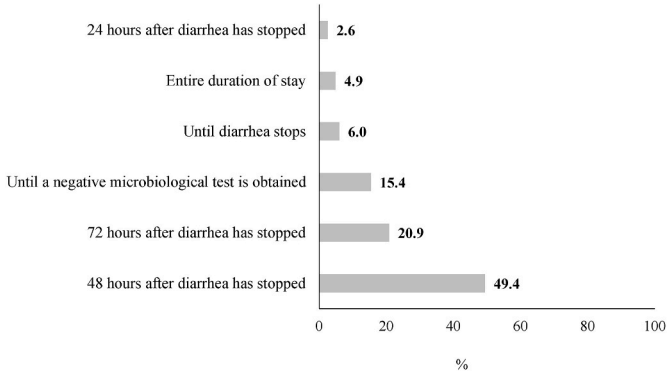


Fig. 3. Duration of contact precautions in hospitals that participated in the national survey.

4. Discussion

While healthcare systems increasingly adopt innovative technologies to reduce health care-associated infections, it is important not to overlook foundational IPC principles such as hand hygiene, isolation precautions, the use of personal protective equipment, and environmental disinfection. In this context, our nationwide survey provides interesting insights into current CDI-related IPC practices across French hospitals.

With broad participation of HCWs – mainly by infection control team members from public hospitals – this study illustrates the variability and implementation challenges of applying international IPC guidelines in real-world clinical settings. Although some key components of CDI prevention – such as daily cleaning of patient environments and hand hygiene – seem to be well-integrated into practices, several gaps were identified. Notably, while 85 % of respondents reported appropriate hand hygiene using soap and water followed by alcohol-based hand rub, only 71.4 % reported using sporicidal agents for environmental cleaning, which are essential for eliminating *C. difficile* spores [8]. Improving environmental disinfection has been shown to positively reduce health-care associated CDI risk [11–14]. Our results are consistent with previous studies showing that even in high-resource settings, adherence to sporicidal cleaning protocols may be hindered by logistic constraints, unclear procedures, or insufficient staff training [15,16].

The suboptimal use of personal protective equipment before entering infected patient rooms may increase the risk of indirect transmission, despite evidence supporting its effectiveness in endemic settings [17].

The application of CP varied widely. While 70 % of respondents reported implementing CP for symptomatic patients, only 49 % adhered to the recommended minimum duration of 48 h after symptom resolution [9]. Interestingly, a small subset (4.9 %) continued CP until patient discharge, suggesting either an overcautious approach or operational constraints. Moreover, 15 % of respondents indicated that a negative test is necessary before stopping CP, reflecting a common misconception in clinical practice. Guidelines specify that isolation should be based on symptom resolution rather than repeat testing. Extending isolation until a negative stool test can unnecessarily consume laboratory resources and prolong patient isolation without additional benefit for infection control. These discrepancies may reflect variable interpretation of guidelines, limited protocol standardization, or workload-related implementation challenges.

Strikingly, none of the participants achieved the full global compliance score defined in the study. Although nearly half (48.3 %) showed "good" adherence, not a single participant reported implementing all recommended measures. This underscores the complexity of translating comprehensive IPC bundles in real-life hospital environments. Previous European studies similarly highlight difficulties in achieving full CDI IPC compliance, even in facilities with established infection control programs [18,19].

The results of this survey highlight the need for targeted interventions to strengthen CDI IPC. Standardizing intuitional protocols, improving staff training, and reinforcing surveillance and feedback mechanisms could help reduce practice variability. Additionally, hospital leadership must ensure that IPC teams have sufficient resources, HCW have access to appropriate disinfectants, isolation capacity, and time for staff training.

This study has some limitations. First, self-reported data may introduce bias, as participants might overestimate their compliance with IPC measures. Second, participants were mainly infection control professionals, which could overestimate adherence levels compared to frontline clinical staff. Finally, the absence of observational data limits the ability to verify actual practices. Although no participant achieved perfect global compliance, analyzing those with higher adherence highlights practices that are more consistently applied and may serve as targets for training and intervention. This insight could guide future initiatives to improve IPC compliance across healthcare settings.

5. Conclusion

Despite high levels of awareness among HCW, this study reveals that full adherence to CDI IPC guidelines remains suboptimal in French hospitals. The results emphasize the need for renewed efforts to translate recommendations into practice through clear protocols, ongoing training, and institutional support. Improving IPC compliance is essential to reducing the burden of *C. difficile* infection and preventing its transmission in healthcare settings.

CRediT authorship contribution statement

Nagham Khanafer: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Anne-Gaëlle Venier:** Writing – review & editing, Methodology. **Raymond Nasso:** Writing – review & editing. **Romane Baroux:** Writing – review & editing, Software, Project administration. **Philippe Vanhems:** Writing – review & editing, Methodology.

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Transparency declaration

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Khanafer reports a relationship with Becton Dickinson that includes: consulting or advisory, funding grants, speaking and lecture fees, and travel reimbursement. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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